Derek David

Senior Full-Stack & AI Systems Engineer

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Professional Summary

Senior Full-Stack & AI Systems Engineer with 5+ years of experience architecting high-concurrency web, mobile, fintech, and voice AI platforms. Specialized in the TypeScript, React, Next.js, React Native, and Node.js ecosystem, with deep production expertise in atomic financial ledgers, real-time WebSocket audio streaming, LLM function calling, and distributed BullMQ/Redis workflows.

Technical Skills

Frontend: JavaScript (ES6+), TypeScript, React.js (React 19), Next.js (App Router), React Native (Expo SDK 54), Kotlin (Android IME), Flutter, Redux Toolkit, TanStack Query, ReactFlow, Vue.js, Tailwind CSS, Shadcn UI / Radix UI
Backend: Node.js, Express.js (v5), RESTful APIs, GraphQL, WebSockets (Socket.io / ws), BullMQ, Microservices, Idempotency & Atomic Ledgers, HMAC Cryptographic Webhooks, Twilio / Plivo / SIP Trunking
AI & Fintech: Google Gemini API (Function Calling), OpenAI Realtime & Agents SDK, ElevenLabs Conversational AI, RAG & Vector Chunking, Kobo-Normalized Double-Entry Ledgers, BaaS Integration (Rubies API), Multi-Gateway Billing (Stripe, Paystack, PayPal)
Databases & Cloud: PostgreSQL (Prisma & Drizzle ORM), Redis, MongoDB, MySQL, AWS, Azure, Docker, Kubernetes, Nginx, CI/CD

Experience

Full-Stack & AI Systems Engineer **Aug 2025 – Present**
Praix Development LTD *Remote*

- Architected PraixAI (AgentLabs Core), a multi-tenant Voice AI & Bulk Calling SaaS integrating ElevenLabs Conversational AI and OpenAI Realtime APIs with Twilio, Plivo, and SIP trunks over bidirectional WebSocket audio streams.
- Engineered a ReactFlow drag-and-drop conversation builder and backend state-machine compiler that translates visual node graphs into deterministic LLM instructions and live mid-call webhook tools (RAG lookup, appointment booking, CRM capture).
- Spearheaded the Dora Fintech ecosystem across a 5-project monorepo (Node.js/Express 5, React Native/Expo, Next.js 16, and a native Kotlin Android IME keyboard), enabling in-keyboard P2P and bank transfers via Rubies BaaS.
- Designed a fault-tolerant financial pipeline featuring Kobo-normalized integer ledgers, SHA-256 request-hashing idempotency middleware, 4-tier KYC limit enforcement, and 5-minute BullMQ reconciliation sweep workers.
- Built a multi-turn agentic banking assistant powered by Google Gemini 3.1 Flash Lite with 6 function-calling tools and bcrypt PIN validation, alongside a 5-gateway billing engine (Stripe, Paystack, PayPal, Razorpay, MercadoPago) using PostgreSQL advisory locks.

Full-Stack Developer **Feb 2025 – Aug 2025**
Connectivo AB *Göteborg, Sweden*

- Architected web applications using React and Vue.js that boosted user engagement by 30%.
- Streamlined data flow between frontend and backend, resulting in 28% faster application performance.
- Enhanced user experience through intuitive UI/UX redesigns, increasing retention by 22%.

Full-Stack Developer **Dec 2022 – Aug 2025**
Sky Enterprise *Yola, Nigeria*

- Engineered scalable web applications using the MERN stack (MongoDB, Express, React, Node), reducing manual business workload by 35%.
- Constructed RESTful APIs that improved data integrity by 25% and reduced response times by 30%.
- Implemented hybrid SQL/NoSQL architecture improving query performance by 40%.
- Deployed microservices using Docker, reducing deployment time by 50% with 99.9% uptime.

Node.js Developer **Jan 2021 – Dec 2022**
Manaknight Inc. *Toronto, Canada*

- Developed backend systems for mobile applications using Node.js and PostgreSQL, acquiring 15,000+ users.
- Built a feature-rich golf game management application that increased user engagement by 45%.
- Designed normalized database schemas that improved data retrieval speeds by 35%.
- Integrated third-party APIs that expanded application functionality while reducing development time by 20%.

Key Projects

Dora – Multi-Channel AI Fintech & Digital Banking Ecosystem | *Node.js (Express 5), TypeScript, React Native (Expo SDK 54), Next.js 16 (React 19), PostgreSQL (Prisma), Redis & BullMQ, Kotlin (Android IME), Google Gemini AI, Socket.io* **Jan 2026 – Oct 2026**

- Engineered a fault-tolerant Node.js/Express 5 and PostgreSQL (Prisma) banking ledger using Kobo-normalized integer math, deduct-first balance patterns, and atomic database transactions ($transaction) for P2P and inter-bank settlements via Rubies BaaS.
- Implemented a SHA-256 request-hashing idempotency engine (IdempotencyKey), 4-tier KYC limit enforcement (NIN/BVN), HMAC-SHA512 webhook verification, and a 5-minute BullMQ reconciliation sweep worker to eliminate double-spending and auto-refund failed transfers.

- Built Dora AI, a conversational banking agent powered by Gemini 3.1 Flash Lite with a 5-turn function-calling loop (execute_transfer, purchase_vas, resolve_account) secured by bcrypt 6-digit PIN verification.
- Developed a native Android Input Method Service (IME) keyboard in Kotlin packaged as a Flutter plugin, managing a 15-state UI machine that allows users to send bank/P2P transfers and buy airtime directly inside third-party messaging apps.
- Constructed MFA-protected Admin and Customer Support web portals using Next.js 16 (React 19, Tailwind v4), featuring PII-masked KYC reviews, double-entry atomic refunds, immutable audit logs, and real-time Socket.io ticket chat.

PraixAI (AgentLabs Core) – Multi-Tenant Voice AI & Telephony SaaS | *TypeScript, Node.js, React 18, PostgreSQL (Drizzle ORM), ElevenLabs AI, OpenAI Realtime API, Twilio & Plivo, WebSockets, ReactFlow* **Aug 2025 – Oct 2026**

- Architected multi-engine telephony and voice pipelines bridging Twilio, Plivo, and native SIP trunks with ElevenLabs Conversational AI and OpenAI Realtime APIs over low-latency bidirectional WebSockets.
- Built a ReactFlow visual agent builder and state-machine compiler supporting live mid-call webhook tool execution for RAG knowledge retrieval, real-time appointment scheduling, and automated post-call CRM lead scoring.
- Engineered a 79-table PostgreSQL schema (Drizzle ORM + Zod) with advisory-locked credit wallets, a 5-gateway payment engine (Stripe, Paystack, PayPal, Razorpay, MercadoPago), and a crash-recoverable bulk campaign job scheduler.

Resume Pro | *Next.js, TypeScript, PostgreSQL (Prisma), Redis & BullMQ, Google Gemini API*

- Architected an AI-powered resume optimization platform using Next.js, PostgreSQL (Prisma), BullMQ, and the Google Gemini API for asynchronous CV parsing, real-time ATS scoring, and automated formatting.

Haven & Hearth | *React, Node.js, TypeScript, PostgreSQL, Prisma*

- Architected a full-stack real estate platform with Node.js, TypeScript, PostgreSQL (Prisma), and React, featuring advanced geospatial property filtering and interactive map integrations.

derekcodes.online | *DNS, SSL, Nginx, Web Infrastructure*

- Deployed and maintained personal portfolio infrastructure with custom DNS routing, Nginx reverse proxy, SSL configurations, and optimized asset delivery.

EDUCATION

Gombe State University — **Nigeria**
Master of Science in Computer Science — *Apr 2026 – 2027*

Gombe State University — **Nigeria**
Bachelor of Science in Computer Science — *Aug 2017 – Aug 2021*